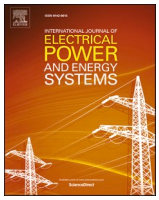
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Influence of Ultra-High-Voltage hybrid reactive power compensation on interruption performance of circuit breakers under different faults

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ABSTRACT

For extremely large ultra-high-voltage (UHV) power transmission and distribution lines with typical project using hybrid reactive power compensation (HRPC), single-phase earth fault (1LG), three-phase ground fault (3LG), phase-to-phase fault (2LS), and two-phase grounding fault (2LG) often occurs. Circuit breaker is one of the most critical equipment of UHV transmission system. And the interruption performance is crucial to circuit breakers' breaking capability. In this study, formal expressions of the indicators for the interruption performance of the circuit breaker are deduced in case of different faults based on the equivalent-valued parameter circuit based on the structure a UHV demonstration project. The simulations are used to validate the accuracy of formal expressions. Then, the influence of such factors as the location of the fault and the degree of HRPC on the interruption performance of the 1LG, 2LG, 3LG, and 2LS is analyzed. Finally, the interruption performance of circuit breakers under different faults are compared. A quantitative evaluation of the influence of changes in the relevant indicators on the interruption performance of circuit breakers can provide a technical basis for choosing the appropriate design of the circuit breaker in light of the application as well as equipment development and assessment.

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1. Introduction

Ultra-high-voltage (UHV) technology has lately emerged as a suitable choice to resolve the contradiction between energy and development owing to such properties as long transmission distance, large transmission capacity, and low energy loss [1]. With the expansion of the power scale of transmission line and increase in the transmission voltage, fluctuations in the reactive power of the system become more frequent, and the traditional method of power compensation cannot meet the requirements of UHV projects [2]. To satisfy these requirements while limiting excessive voltage in the system, it is necessary to use the hybrid reactive power compensation (HRPC) processing method with series compensation (SC) and a stepped controlled shunt reactor (SCSR) [3]. This method can effectively solve the loopholes in the previous forms of reactive power compensation and improves the flexibility and stability of power compensation in the UHV system [4].

The common UHV power transmission circuit faults can generally be divided into two categories: earth faults and phase-to phase-fault(2LS). The earth fault can be divided into single-phase earth fault (1LG), two-phase grounding fault (2LG), three-phase ground fault (3LG). 3LG is a very serious fault problem, but 1LG is more likely to occur [5]. However, when the 2LS occurs, the insulation between the connections of the set phase is further destroyed, and the network architecture of the power system has changed correspondingly, which causes the change of the power transmitted by the generator and the platform system to be volatile [6]. Therefore, it is of great practical significance to explore the effect of UHV on the interruption performance of circuit breakers (CB) in case of these faults [7].

Recently, a huge amount of research has been carried out on the transient recovery voltage (TRV) and rate of rise of TRV (RRRV) of the UHV transmission system. The actual impact of each capacitor bank structure on the TRV of power system is analysed [8]. The TRV experiment of the actual circuit breaker connected to the capacitor bank in the wind power plant connected to the electric power system has been completed [9]. The computational formula of RRRV was deduced through equivalent lumped parameter circuit [10]. Systematic

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investigations have been carried out on various fault conditions caused by power stations through the corresponding electromagnetic transient program (EMTP). At the same time, a specific analysis of the variable index value, the RRRV and the TRV is formed accordingly [11]. In the previous study, we have studied the interruption performance of the circuit breaker in case of phase-to-phase fault and out-of-phase fault of UHV transmission line with HRPC [12]. Zheng Bin et al. study the circuit breaker TRV issues of UHV high series compensation lines with different faults [13]. Nowadays, most of the studies focus on the TRV with the transmission system installed single type of compensation device, or on the analysis for the interruption performance of the circuit breaker under single type of fault when HRPC is installed. There are few comparative studies on the interruption performance of the circuit breaker under multiple faults of the transmission system installed HRPC. And in most studies, the simulation method is conduct whereas the computation method is rarely adopted. In our study, the calculation method is used to compare the interruption performance of circuit breakers for different faults occur on transmission lines with HRPC, which can provide a feasible technical scheme for reliable switching of circuit breakers for UHV system with HRPC.

The specific structure of this paper is shown below. In section 2, the structure of HRPC is introduced. The derivation of formulae for the RRRV in the event of 1LG is described in Section 3. In Section 4, the influence of HRPC on the interruption performance of the CB in case of the 1LG, 2LG, 2LS, and 3LG is analysed. The interruption performance of CBs in case of the occurrence of 1LG, 2LG, 2LS, and 3LG, is compared in section 5. The results of this study provide a theoretical basis for the future application and promotion of HPRC and the design of circuit breakers in UHV transmission lines.

2. Model of a UHV transmission system with HRPC

2.1. Introduction to HRPC

HRPC consists of series compensation (SC) and a stepped controllable shunt reactor (SCSR). The structure is shown in Fig. 1 below [14]. As shown in Fig. 1(a), the SC device is consisted of bypass isolating switch, series isolating switch, capacitor, damping switch, metal oxide arrester (MOA), spark gap and bypass circuit breaker. In the SC device, the MOA is the main protection of overvoltage. The trigger gap is the overheat protection to prevent the arrester from exceeding its thermal capacity. The gap arcs can be effectively extinguished by the isolating switch. The isolating switch cooperate with the series and parallel disconnectors to provide conditions for the whole device to be put into and out. When the fault occurs, the SC device bypasses the series compensation capacitor bank [15]. Therefore, the SC device can be equivalent to a capacitor. As shown in Fig. 1(b), the SCSR is composed of transformer, series reactance, mechanical switch and thyristor [16]. The capacity of SCSR is adjusted by controlling the thyristor switch, and the application of mechanical switch simplifies the cooling system of thyristor. Therefore, the SCSR device can be equivalent to a reactor which can be represented by L_R in Fig. 2[17].

In the series capacitor compensation device, the capacitor bank is the main component of the series compensation device, and the MOV is the main protection device of the series compensation. It is mainly used to quickly disconnect the capacitor bank in case of overvoltage, the gap is used for backup protection, and the damping device is used to prevent damage to the internal equipment of the series compensation when the transient current of the capacitor bank inside the series compensation is too large.

The main part of SCSR is a high leakage reactance transformer structure. The voltage reduction effect of the transformer is utilized to enable the thyristor to work at low voltage. It is used to adjust the leakage reactance of the transformer to make the leakage reactance value close to 100 % rated impedance. The secondary side of the high impedance transformer is composed of four series reactance, which are controlled by bidirectional thyristors connected in parallel on both sides to switch on and off. The mechanical switch is used to bypass the thyristors, achieving the regulation of the inductive reactance capacity when integrated into the power grid.

2.2. UHV transmission system

The equivalent model of the power system for Jindongnan-Nanyang-Jingmen UHV AC typical project is shown in Fig. 2 below [18]. SCs are set at both terminals of the power system close to the circuit side, and SCSRs are set at the terminal close to the power source side of the system. The corresponding physical index parameters and line performance of transmission and distribution lines are shown in Table 1[19].

In Fig. 2, U_m and U_n are the corresponding high voltage power supplies, L_s is the internal inductance of the power supply, C_s is its earth capacitance, L_R is the shunt reactance of the SCSR, L_n is its neutral reactance, C_c is SC capacitance, C_m is the mutual capacitance of the normal and the fault phases, and Z represents the impedance of the transmission line.

3. Calculating interruption performance of a single-phase earth fault

To explore the interruption performance of the circuit breaker, a quantitative assessment is set to confirm its index factors. The calculation method is selected. Compared with the traditional simulation method to analyze the interrupting characteristics of circuit breaker, the numerical calculation is quickly, and can well analyze the influence of line parameters on the interrupting characteristics of circuit breaker. A quantitative evaluation of the influence of changes in the relevant indicators on the interruption performance of circuit breakers can provide a technical basis for choosing the appropriate design of the circuit breaker in light of the application as well as equipment development and assessment. In various operating environments, factors such as the maximum value of TRV, the rate rise of the transient recovery voltage (RRRV), and the drop-out current is set as the interruption performance in different ways.

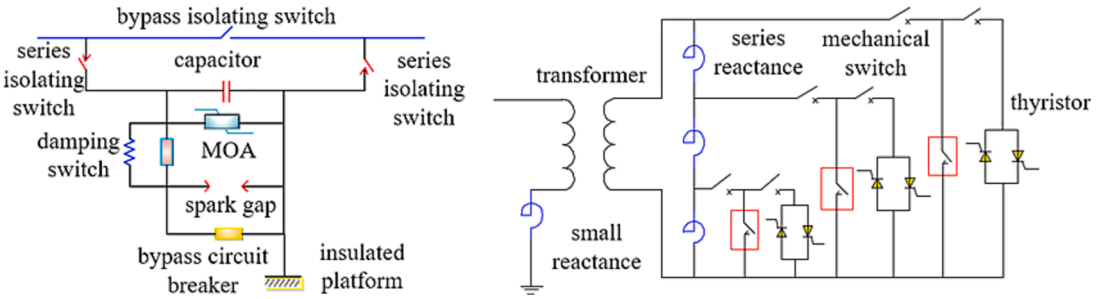


Fig. 1. Structural principle of HRPC (a) SC; (b) SCSR.

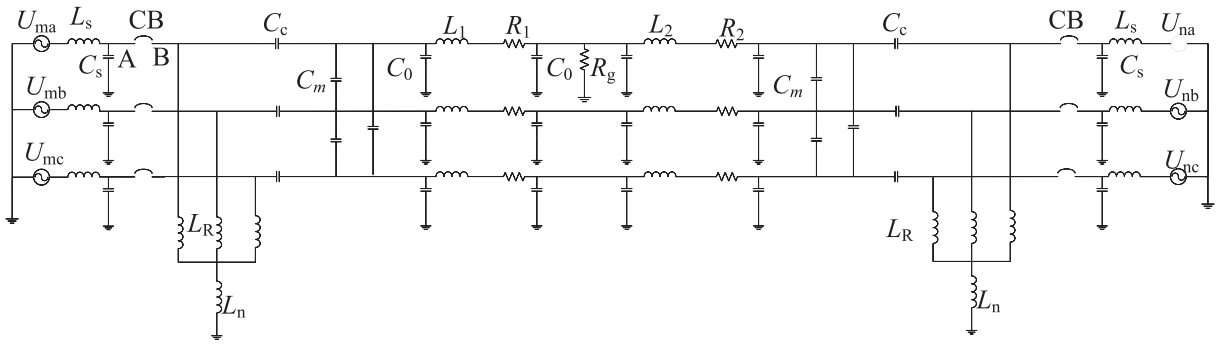


Fig. 2. Equivalent circuit of power system.

Table 1
Core parameters of 1000 kV transmission circuit.

	Parameter	Value
Numerical parameters of transmission line (positive sequence / zero sequence)	System frequency	50 Hz
	System power bus level	1100 kV
	Short circuit capacity between three phases	50 GVA
	Line length	645 km
	R	0.00783 Ω·km^{-1}
	L	0.19637 Ω·km^{-1}
		0.83408 mH·km^{-1}
	C	2.67366 mH·km^{-1}
		0.01379 μF·km^{-1}
		0.00846 μF·km^{-1}

3.1. The main variable elements that affect the shutdown performance of single-phase earth fault

The severity F of CB can be calculated according to the maximum value of TRV, RRRV and drop-out current [20]:

$$F = U_{\text{tr max}} \cdot (RRRV)^{\alpha} \cdot I_{\text{ss}} \tag{1}$$

α is a constant, which is caused by the arc extinguishing form of the CB and the arc extinguishing medium, which is troublesome to calculate.

3.2. Influence of HRPC on interruption performance of single-phase earth fault

In the UHV system, the assembly of HRPC will affect the interruption performance. In order to explain the difference for the interruption performance of the 1LG in the UHV power system with and without HRPC. It is vital to analyze the drop-out current and TRV of the CB under these two types of conditions. According to the international standards [21], the definition of drop out current in the circuit breaker is the current in the pole of circuit breaker at the instant of arc initiation during a breaking process.

Owing that the ground branch current is small, it is ignored in Fig. 2. Thus, the drop-out current of the 1LG, I_{op}, can then be approximated as

$$\begin{aligned} I\text{op} &= \frac{U_{\text{p}}}{\sqrt{\left(\omega L_s - \frac{1}{\omega C_c} + \omega L_1\right)^2 + R_{\text{g}}^2}} + \frac{U_{\text{p}}}{\sqrt{\left(-\frac{1}{\omega C_{\text{m}}}\right)^2}} \\ &= \frac{U_{\text{p}}}{\sqrt{\left(\omega L_s - K_c \omega L_1 + \omega L_1\right)^2 + R_{\text{g}}^2}} + \frac{U_{\text{p}}}{\sqrt{\left(-\frac{1}{\omega C_m}\right)^2}} \end{aligned} \tag{2}$$

In the expression, L_1 is the reactance of the circuit, and R_g is the arc resistance. The compensation degree of SC and SCSR (K_c and K_R) can be set to β and γ which in turn are determined by

$$\beta = \frac{X_{C_c}}{X_{L_1}} = \frac{\left(\frac{1}{\omega C_c}\right)}{\omega L_1} = \frac{1}{\omega^2 L_1 C_c} \tag{3}$$

$$\gamma = \frac{X_{L_R}}{X_{C_1}} = \frac{\omega L_R}{\left(\frac{1}{\omega C_1}\right)} = \omega^2 L_R C_1 \tag{4}$$

where C_1 is line capacitance, X_{Cc}, X_{L1}, X_{LR}, and X_{C1} are the complex impedances of C_c, L_1, L_R, and C_1, respectively.

The oscillation periods of the voltage on the power supply side T_1, the SC voltage T_2, and the line voltage T_3, are respectively given by

$$T_1 = 2\pi\sqrt{L_s C_s} \tag{5}$$

$$T_2 = 2\pi\sqrt{L_R C_c} \tag{6}$$

$$T_3 = 2s\sqrt{l_1 c_1} \tag{7}$$

In the expression, l_1 and c_1 are the reactance and capacitance corresponding to unit length.

The starting oscillation time of TRV, T_0, is confirmed by the lowest oscillation period value between the power supply terminal SC and the line level, i.e., it is given by the smallest value among T_1, T_2, and T_3, and is denoted by min (T_1, T_2, T_3). The results of calculation show that min (T_1, T_2, T_3) is equal to T_1, and thus

$$t_{\text{p}} = \frac{T_0}{2} = \frac{\min(T_1, T_2, T_3)}{2} = \frac{T_1}{2} = \pi\sqrt{L_s C_s} \tag{8}$$

Here, t_{p} is the time required for the recovery level to reach the first maximum value.

Furthermore, the oscillation period of the initial TRV is equal to the minimum value of that of the voltage on the power supply side U_A. The line loss is ignored during the initial transmission of the traveling wave. During the 0-T3/2 period, the circuit level changes are shown as follows:

$$\frac{du_{\text{L}}}{dt} = \frac{2U_{Lm}}{s\sqrt{l_1 c_1}} = 2I_{\text{op}}(\omega l_1 s)\frac{1}{s\sqrt{l_1 c_1}} = 2I_{\text{op}}\omega Z_0 \tag{9}$$

here, s is the fault distance, Z_0 and U_{Lm} are the wave impedance and the maximum voltage on the transmission line, and can be expressed as

$$Z_0 = \sqrt{\frac{l_0}{c_0}} \tag{10}$$

$$U_{Lm} = I_{\text{op}}(\omega l_1 s) \tag{11}$$

$$U_{\text{L}} = U_{\text{Lm}} - I_{\text{op}}\omega Z_0 t \tag{12}$$

Typically, the process of interruption of the CB occurs at the zero-

crossing of the fault current. After interruption by the 1LG, the TRV between the port of CB_1 is calculated as

$$\begin{aligned} U_{\mathrm{tr}} &= u_{\mathrm{M}} - u_{\mathrm{N}} \\ &= U_{\mathrm{m}}\cos 100\pi t + (U_{\mathrm{m}} \\ &\quad - U_{\mathrm{Mm}})\cos\left(\frac{1}{\sqrt{L_{\mathrm{s}}C_{\mathrm{s}}}}\right)t - U\mathrm{Hm} - I_{\mathrm{op}}\omega(l_1 s) + I_{\mathrm{op}}\omega Z_0 t \end{aligned} \tag{13}$$

here, U_{Mm} is the maximum value of the level in the CB_1 port M, U_{Hm} is the amplitude of the voltage on the SC after interruption by the 1LG. They are given by

$$U_{\mathrm{Am}} = I_{\mathrm{op}}\omega L_s \tag{14}$$

$$U_{\mathrm{Hm}} = u_{C_{\mathrm{c}}}\cos\omega_2 t = \frac{I_{\mathrm{op}}\cos\omega_2 t}{\omega C_{\mathrm{c}}} \tag{15}$$

The angular frequency of the oscillation in voltage on the SC is

$$\omega_2 = \frac{1}{\sqrt{L_{\mathrm{R}}C_{\mathrm{c}}}} = \frac{1}{\sqrt{\omega^4\beta\lambda l_1 c_1 s^2}} \tag{16}$$

From (1) and (10)– (15), the RRRV of CB_1 in the event of 1LG can be calculated as follows:

$$\begin{aligned} RRRV &= \frac{u_{\mathrm{tr}}|_{t=t_{\mathrm{p}}/2} - u_{\mathrm{tr}}|_{t=t_0}}{\frac{1}{2}t_{\mathrm{p}}} = \frac{u_{tr}|_{t=t_{\mathrm{p}}/2} - u_{\mathrm{tr}}|_{t=t_0}}{\frac{1}{2}\pi\sqrt{l_{\mathrm{s}}c_{\mathrm{s}}}} \\ &= \frac{2}{\pi\sqrt{L_{\mathrm{s}}C_{\mathrm{s}}}}\left[U_{\mathrm{m}}\cos 100\pi\left(\frac{1}{2}t_{\mathrm{p}} - t_0\right) + (U_{\mathrm{m}} - U_{\mathrm{Am}})\cos\left(\frac{1}{\sqrt{L_{\mathrm{s}}C_{\mathrm{s}}}}\right)\left(\frac{1}{2}t_{\mathrm{p}} - t_0\right)\right. \\ &\quad \left. - U_{\mathrm{Cc}}\cos\omega_2\left(\frac{1}{2}t_{\mathrm{p}} - t_0\right) - I_{\mathrm{op}}\omega(l_1 s) + I_{\mathrm{op}}\omega Z_0\left(\frac{1}{2}t_{\mathrm{p}} - t_0\right)\right] \end{aligned} \tag{17}$$

A comparison between the results of calculations and the simulation of the TRV is shown in Fig. 3, from which are consistent.

Fig. 3 Comparison between the results of calculation of the TRV and those of the simulation cause a phase to ground fault.

After substituting the corresponding variable values in Table 1 into expression (17), the RRRV is 4.8 kV/μs. The value obtained by the simulation is 4.87 kV/μs, which is same with the calculation result.

The factors influencing the interruption performance of the CB in case of 1LG, which include the drop-out current, the first maximum value of the TRV and RRRV—with the degree of SC and SCSR are shown in Figs. 4, 5, and 6,respectively.

Fig. 4 shows that the drop-out current fluctuated with the increase in SC, and its maximum value occurred when $K_{\mathrm{c}} = 0.4$ and the minimum

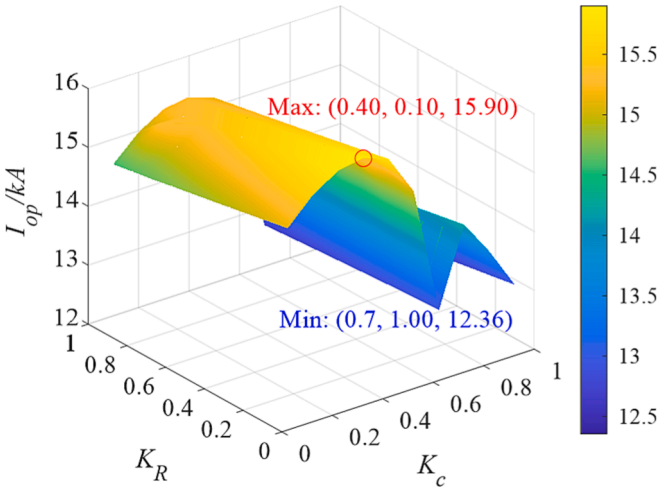


Fig. 4. Combined variations in drop-out current with degrees of compensation of the SC and SCSR in case of a single-phase-to-ground fault.

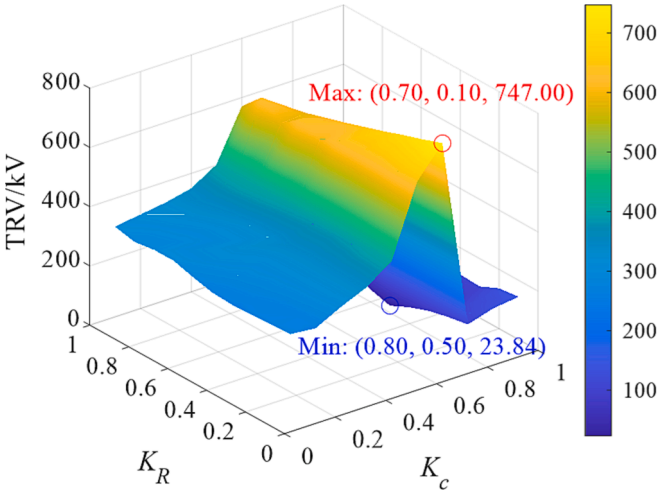


Fig. 5. Under the condition of a connected ground fault and damage, TRV and SC and SCSR compensation degree reorganized and changed.

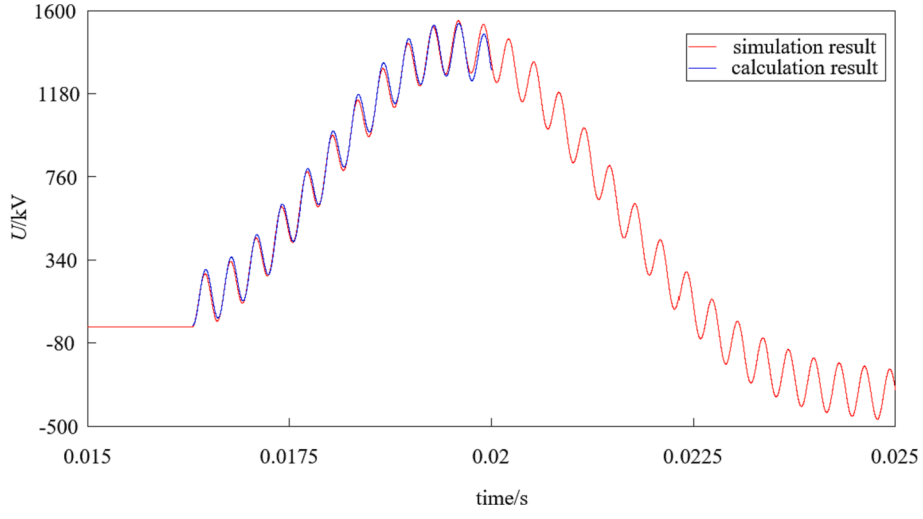


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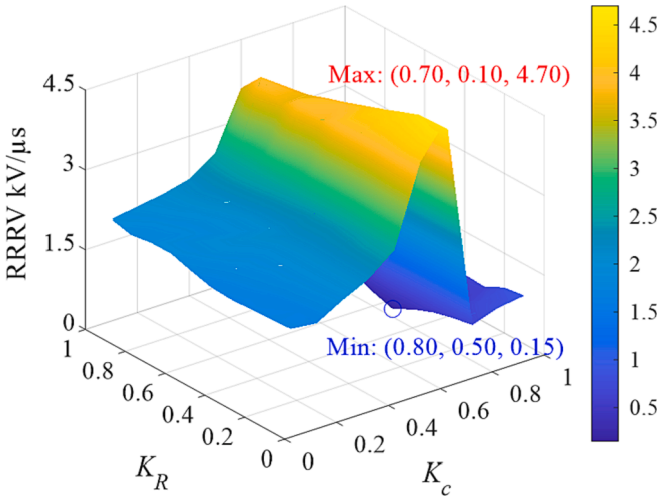


Fig. 6. Combined variation of RRRV with the degrees of compensation of SC and SCSR in case of 1LG.

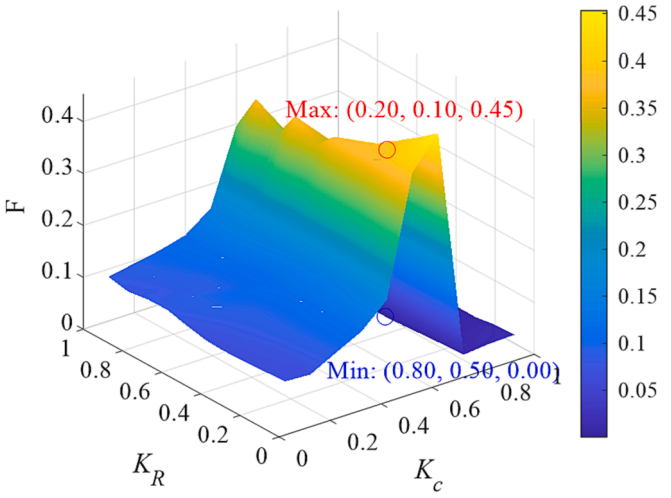


Fig. 7. Combined variation in the severity when α = 1 with degrees of compensation of the SC and SCSR in case of 1LG.

value when $K_c = 0.7$. The drop-out current remained unchanged with the increase in K_R.

Fig. 5 shows the variation of TRV and the degree of SC and SCSR. From the results, it can be found that TRV increases first and then decreases with the increase in the SC compensation level, and reaches the highest value when $K_c = 0.7$, but it remains fixed with the increase of the SCSR compensation level. Therefore, TRV continues to increase with the assembly of HRPC. The higher the SC degree and the smaller the series capacitance is, and the higher the natural oscillation frequency of the fault circuit of SC line. It causes the TRV waveform peak time to be shorter and the RRRV to be higher, thus affecting the TRV and RRRV of the circuit breaker, and ultimately affecting the interruption performance of the circuit breaker.

In the previous study [15], China State Grid Corporation Company (SGCC) carried out supplementary test research for the TRV of the circuit breakers of UHV AC series compensation lines. The research results in the dissertation show that the peak value of the TRV of circuit breaker is 532 kV when 1LG occurs in the 1100 kV UHV transmission line with the 40 % series compensation. The peak value of our calculation result in this manuscript is 536 kV which is consistent with the results in our paper within a very small margin of error. Therefore, the validity of the proposed method can be confirmed indirectly.

Fig. 6 shows the RRRV changes in the degree of compensation level of SCSR associated with the SC. With the further increase of SC, RRRV tends to increase first and then decrease. The maximum value occurs when $K_c = 0.7$. In contrast, with the increase of the SCSR degree, it always remains constant. Therefore, after the HRPC is assembled, the RRRV shows a tendency to increase first and then decrease.

The value of the alpha α is a constant, which is decided by various parameters, including the extinguishing medium and extinguishing method of the circuit breaker. Thus, the value of α is difficult to calculate due to changes in the extinguishing medium and extinguishing method of the circuit breaker. Therefore, we have analyzed the variation trend of the severity values for the different arc extinguishing constants, such as 1 and 2, and the following results are obtained.

When the arc-extinguishing constant α = 1, the effect of the degree of HRPC compensation on the interruption performance of the CB is explored. Fig. 7 below shows that after HRPC is assembled and 1LG is generated in UHV system, when the SC compensation level is 0.2, the CB will be more difficult to interrupt, and the interruption conditions becomes more severe. The compensation level of SCSR has a relatively small effect on the level of disconnection. In addition, after the HRPC is set on the UHV transmission circuit, the drop-out current, TRV, and RRRV of CB will show a significant increase, which increased the severity of interruption of 1LG and eventually increased the probability of failure of the CB.

When the arc-extinguishing constant α = 2 the variation trend of the severity is the same as the trend when the arc-extinguishing constant α is taken as 1, but the value and the maximum value is larger than the value when the arc-extinguishing constant α is taken as 1, which indicates that when the arc extinguishing method and arc extinguishing medium of the circuit breaker are more affected by the RRRV(the higher value of arc-extinguishing constant α), the value of the severity will be higher. Therefore, if the arc extinguishing method and extinguishing medium of the circuit breaker are more affected by the RRRV in actual situations, the magnitude of the RRRV should be limited as much as possible to make the arc more easily extinguished and the circuit breaker be opened smoothly.

4. Quantitative evaluation of interrupting performance of UHV HRPC circuit breakers under different faults

4.1. Influence of HRPC on interruption performance of three-phase ground faults

The method proposed in Section 3.2 was used to calculate the interruption performance of a UHV transmission line in case of 3LG, and the index of interruption performance was obtained as:

$$Iop = \frac{U_p}{\sqrt{\left(\omega L_s - \frac{1}{\omega C_c} + \omega L_1\right)^2 + R_g^2}} \tag{18}$$

$$\begin{aligned} U_{tr} &= u_M - u_N \\ &= U_m \cos 100\pi t + \left(U_m - I_{op}\omega L_s\right)\cos\left(\frac{1}{\sqrt{L_s C_s}}\right)t - \frac{I_{op}\cos\frac{t}{\sqrt{\omega^4\beta\gamma l_1 c_1 s^2}}}{\omega C_c} \\ &\quad - I_{op}(\omega l_1 s) + I_{op}\omega Z_0 t \end{aligned} \tag{19}$$

$$\mathrm{RRRV}=\frac{u_{tr}|_{t=t_p/2}-u_{tr}|_{t=t_0}}{\frac{1}{2}\pi\sqrt{l_s c_s}}$$
$$=\frac{2}{\pi\sqrt{L_s C_s}}\left[U_m\cos 100\pi\left(\frac{1}{2}t_p-t_0\right)+(U_m-U_{Am})\cos\left(\frac{1}{\sqrt{L_s C_s}}\right)\left(\frac{1}{2}t_p-t_0\right)\right.$$
$$\left.-U_{Cc}\cos\omega_2\left(\frac{1}{2}t_p-t_0\right)-I_{op}\omega(l_1 s)+I_{op}\omega Z_0\left(\frac{1}{2}t_p-t_0\right)\right] \quad (20)$$

After the corresponding parameter variable in Table 1 taking into expression (20), it can be known that the RRRV is 2.12 kV/μs. The RRRV obtained by the simulation is 2.14 kV/μs, which remains the same as the result of the calculation.

To analyze the influence of the degree for the compensation of HRPC on the severity of interruption by a CB in case of a 3LG, the arc-extinguishing constant α was set to one to achieve the tendency of variation in the corresponding severity level of interruption. Fig. 8 shows the tendency of severity with HRPC.

Fig. 9(a) shows that when the arc-extinguishing constant α was one, the severity of interruption first increased and then decreased with the increase in K_c. Moreover, if K_c was 0.7, the severity of interruption reached its maximum value when a fault occurred in the head of the transmission line. This fully shows the assumed compensation level of SC was 0.7 and 3LG occurred, the severity of interrupting for the CB was the highest. As is shown in Fig. 9(b), the tendency of variation in different levels of corresponding severity of compensation of the SC was small.

4.2. Influence of HRPC on interruption performance of phase-to-phase faults

The method proposed in Section 3.2 was also used to analyze the interruption performance of a UHV transmission line in case of 2LS. The index of the interruption performance of 2LS was obtained as:

$$I\mathrm{op}=\frac{U_p|\sin(\omega t+\theta_1)-\sin(\omega t+\theta_2)|}{\sqrt{\left(2\omega L_s-\frac{2}{\omega C_c}+\omega L_1\right)^2+(2R_g)^2}}$$
$$=\frac{U_p|\sin(\omega t+\theta_1)-\sin(\omega t+\theta_2)|}{\sqrt{(2\omega L_s-2\beta\omega L_1+\omega L_1)^2+(2R_g)^2}} \quad (21)$$

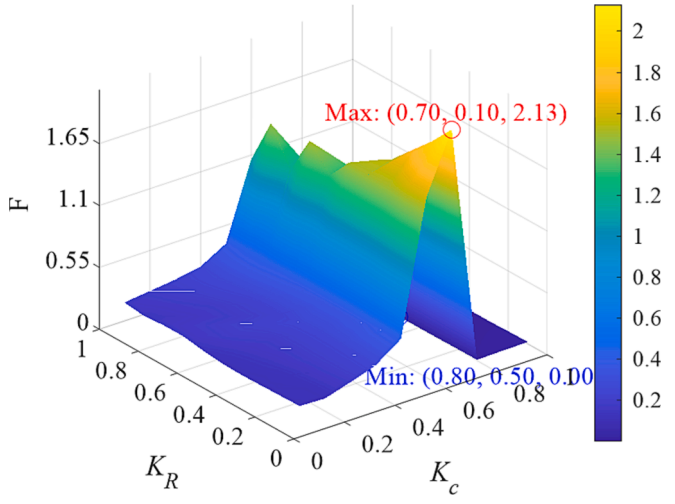


Fig. 8. Combined variation in the severity when α = 2 with degrees of compensation of the SC and SCSR in case of 1LG.

$$U_{tr}=u_M-u_N$$
$$=U_m\cos 100\pi t+(U_m-I_{op}\omega L_s)\cos\left(\frac{1}{\sqrt{L_s C_s}}\right)t-\frac{I_{op}\cos\frac{t}{\sqrt{\omega^4\beta\gamma l_1 c_1 s^2}}}{\omega C_c}$$
$$-I_{op}(\omega l_1 s)+I_{op}\omega Z_0 t \quad (22)$$

$$\mathrm{RRRV}=\frac{u_{tr}|_{t=t_p/2}-u_{tr}|_{t=t_0}}{\frac{1}{2}\pi\sqrt{l_s c_s}}$$
$$=\frac{2}{\pi\sqrt{L_s C_s}}\left[U_m\cos 100\pi\left(\frac{1}{2}t_p-t_0\right)+(U_m-U_{Am})\cos\left(\frac{1}{\sqrt{L_s C_s}}\right)\left(\frac{1}{2}t_p-t_0\right)\right.$$
$$\left.-U_{Cc}\cos\omega_2\left(\frac{1}{2}t_p-t_0\right)-I_{op}\omega(l_1 s)+I_{op}\omega Z_0\left(\frac{1}{2}t_p-t_0\right)\right] \quad (23)$$

Among them,θ is the phase angle of the fault. After substituting the corresponding parameter values in Table 1 into expression (23), the calculated RRRV is 1.75 kV/μs. The RRRV value obtained by the simulation is 1.86 kV/μs, which is the same as the final calculation result.

To research the relationship between the degree of compensation of HRPC and the severity of interrupting by a CB in case of a 2LS, arc extinguishing constant α is assumed as 1 to obtain the changing tendency of the severity level corresponding to the disconnection. Fig. 10 shows the tendency of severity with HRPC at different fault spacing.

As shown in Fig. 10(a) that severity first increased and then decreased as K_c increased. When the distance between faults does not exceed 150 km, the corresponding severity level reaches the highest value when K_c = 0.4. Fig. 10(b) fully explain that with the SCSR compensation level increased, the corresponding severity level remains basically fixed. When the fault problem occurs at a location 150 km away from the platform system, severity was higher than at other distances. The higher the SC degree is, the overvoltage to ground of CB rises. When it is opposite to the CB line side voltage, the TRV amplitude of CB will upgrade.

4.3. Influence of HRPC on interruption performance of two-phase grounding fault

The proposed method was used to analyze the interruption performance of a UHV transmission line in case of 2LG. The index of the interruption performance of 2LG was obtained as:

$$I\mathrm{op}=\frac{U_p|\sin(\omega t+\theta_1)-\sin(\omega t+\theta_2)|}{\sqrt{\left(2\omega L_s-\frac{2}{\omega C_c}+\omega L_1\right)^2+R_g^2}}=\frac{U_p|\sin(\omega t+\theta_1)-\sin(\omega t+\theta_2)|}{\sqrt{(2\omega L_s-2K_c\omega L_1+\omega L_1)^2+R_g^2}} \quad (24)$$

$$U_{tr}=u_M-u_N$$
$$=U_m\cos 100\pi t+(U_m-I_{op}\omega L_s)\cos\left(\frac{1}{\sqrt{L_s C_s}}\right)t-\frac{I_{op}\cos\frac{t}{\sqrt{\omega^4\beta\lambda l_1 c_1 s^2}}}{\omega C_c}$$
$$-I_{op}(\omega l_1 s)+I_{op}\omega Z_0 t \quad (25)$$

$$\mathrm{RRRV}=\frac{u_{tr}|_{t=t_p/2}-u_{tr}|_{t=t_0}}{\frac{1}{2}\pi\sqrt{l_s c_s}}$$
$$=\frac{2}{\pi\sqrt{L_s C_s}}\left[U_m\cos 100\pi\left(\frac{1}{2}t_p-t_0\right)+(U_m-U_{Am})\cos\left(\frac{1}{\sqrt{L_s C_s}}\right)\left(\frac{1}{2}t_p-t_0\right)\right.$$
$$\left.-U_{Cc}\cos\omega_2\left(\frac{1}{2}t_p-t_0\right)-I_{op}\omega(l_1 s)+I_{op}\omega Z_0\left(\frac{1}{2}t_p-t_0\right)\right] \quad (26)$$

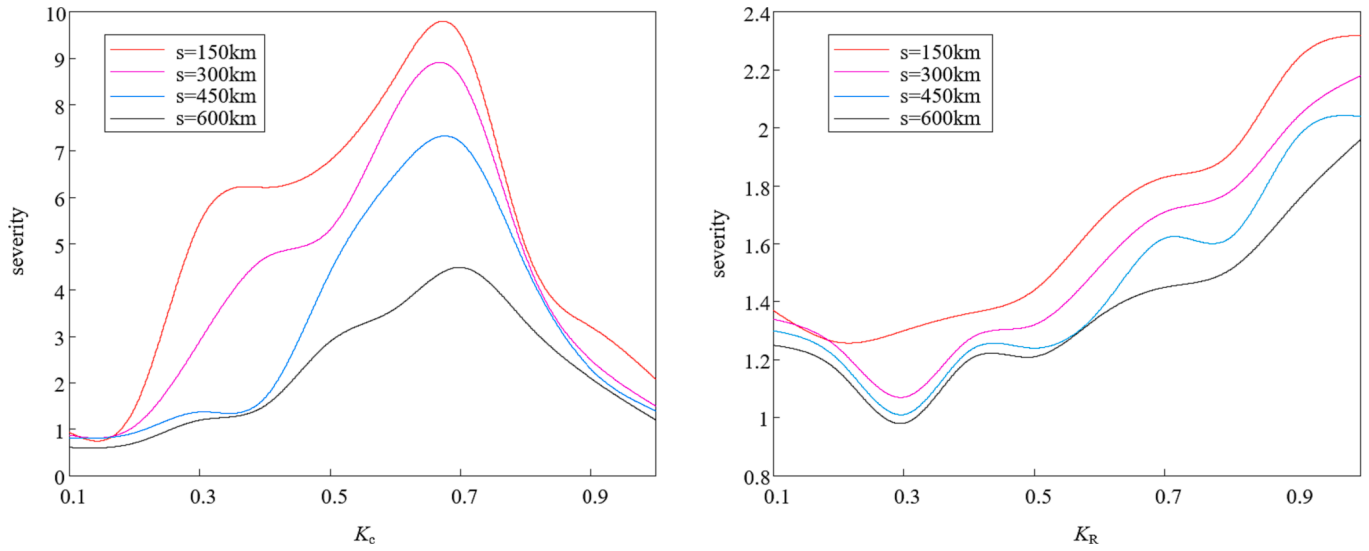


Fig. 9. Variation in the severity of interruption with the degree of compensation in case of 3LG for different fault spacing (a) SCSR; (b) SC.

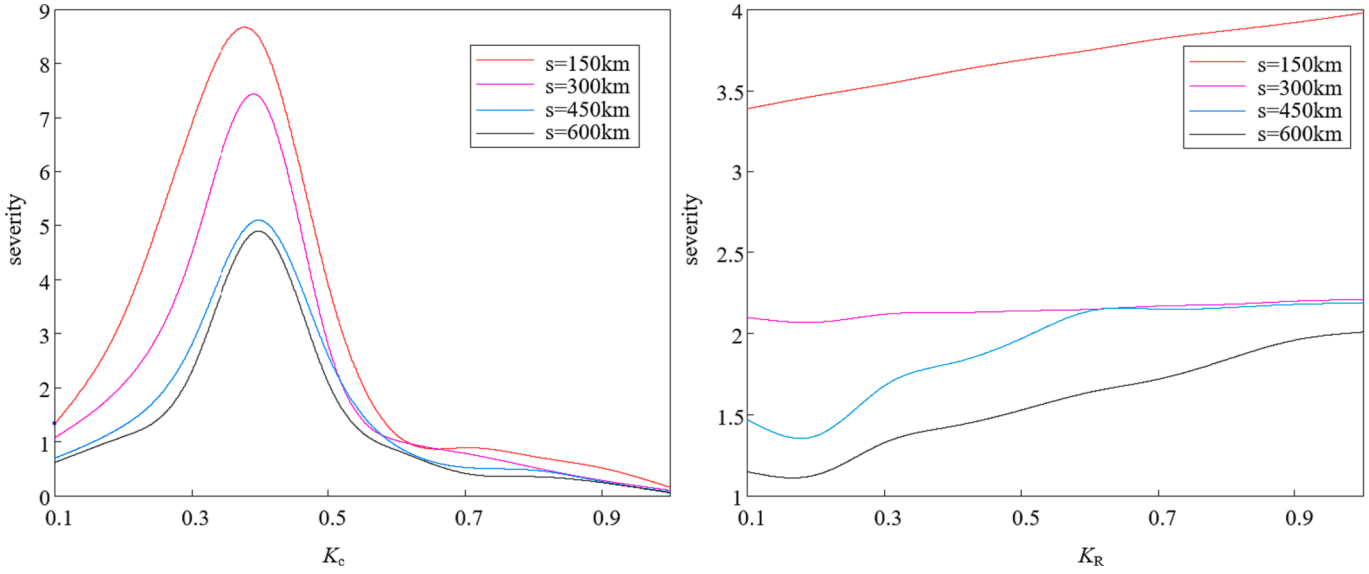


Fig.10. Variation in the severity of interruption with degree of compensation in case of 2LS at different fault distances (a) SCSR; (b) SC.

Substitute the corresponding parameter values in Table 1 into the expression. (26), the RRRV was calculated as 1.65 kV/μs. The RRRV value obtained by the simulation will be 1.76 kV/μs, which remains the same as the result of the operation.

To examine the effect of the degree for the compensation of HRPC on the severity of interruption of CB in case of 2LG, the arc-extinguishing constant $\alpha = 1$ was used to acquire the tendency of variation in the severity of interruption. Fig. 11 shows the corresponding severe level tendency with HRPC in case of 2LG at different fault spacing.

Fig. 11(a) and 11(b) show the tendency of variation in the severity with the degrees of compensation of the SC and SCSR, respectively. Fig. 11(a) shows that the severity first increased and then decreased with the increase in K_c. If K_c was 0.7, the severity of interruption reached its maximum value, which indicates that if the degree of compensation of SC was 0.7, the severity of interruption of the CB was the highest. When a fault occurred at 150 km from the system, the severity was lower than at other spacing. But when the fault spacing was 450 km, in other fault distances, this value is relatively large. Fig. 11(b) below shows that the SCSR compensation level has a relatively small effect on TRV and RRRV, and this provides insight into conditions suitable for the on-site use of the SCSR. The reasons for above phenomenon are that the higher the SC degree is, the larger the current flowing through the SC device, the higher the overvoltage level across the SC is, which makes the overvoltage on the line side to ground rise, resulting in the increase of voltage difference between the two sides of the CB.

5. Comparison between interruption PERFORMANCE of UHV HRPC circuit breakers under different faults

When a fault occurs in the system, the transient performance of the TRV can be decomposed into power frequency components and transient components. As an important indicator that affect the interruption performance of circuit breakers, the TRV depends on the recovery voltage of power frequency, and the distribution of electrical components in the grid loop. When the fault occurs in the system, the distribution of electrical components changes. Once HRPC has been set, the transient performance of the loop become more complex due to the presence of components of inductance and capacitance. The UHV system

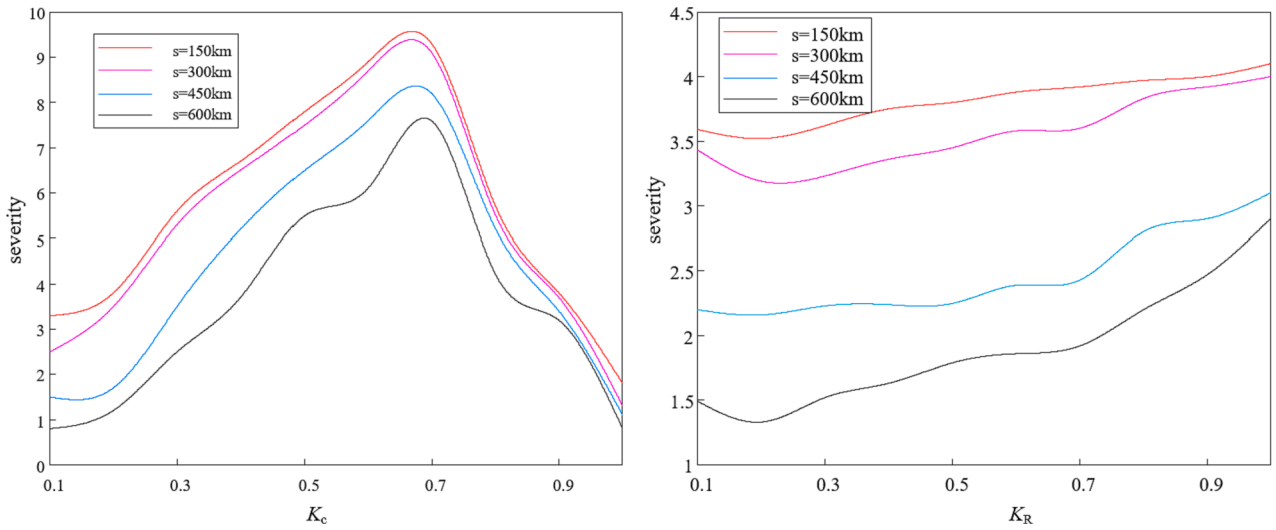


Fig. 11. Variation in the severity of interruption with degree of compensation in case of 2LS at different fault distances (a) SCSR; (b) SC.

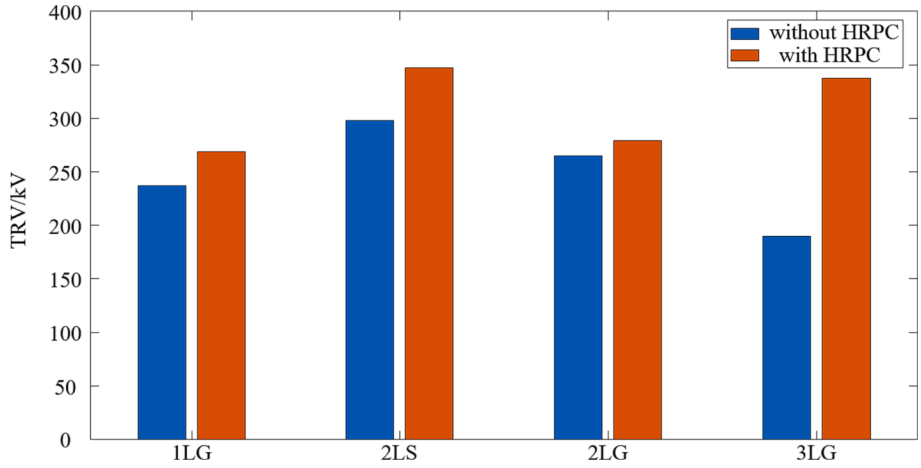


Fig. 12. Comparison of drop-out currents before and after setting HRPC under different faults.

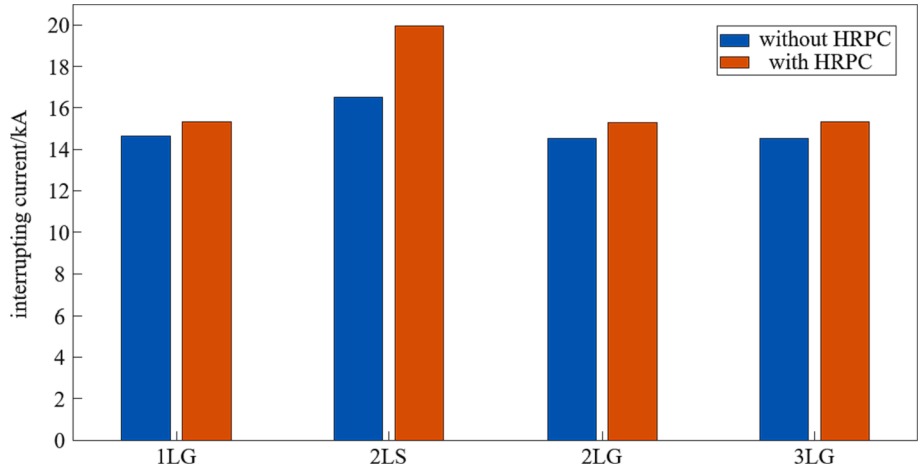


Fig. 13. Comparison of TRVs before and after setting HRPC under different faults.

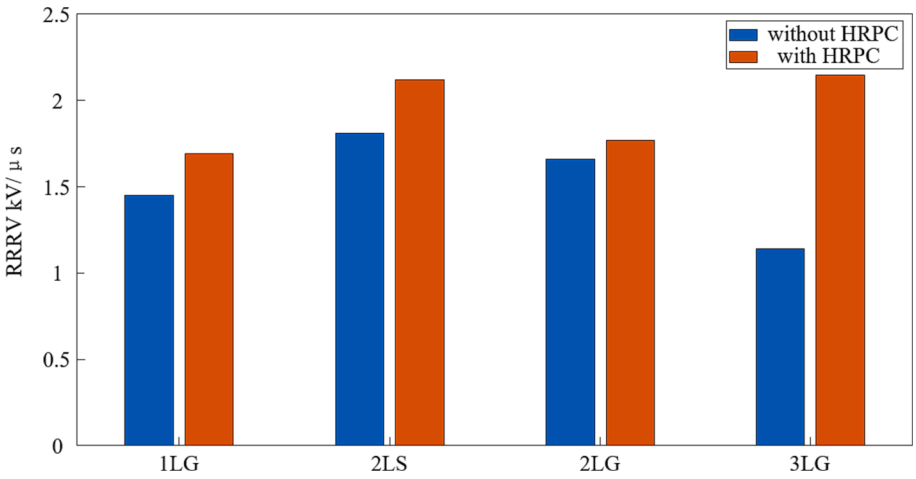


Fig. 14. Comparison of RRRVs before and after setting HRPC under different faults.

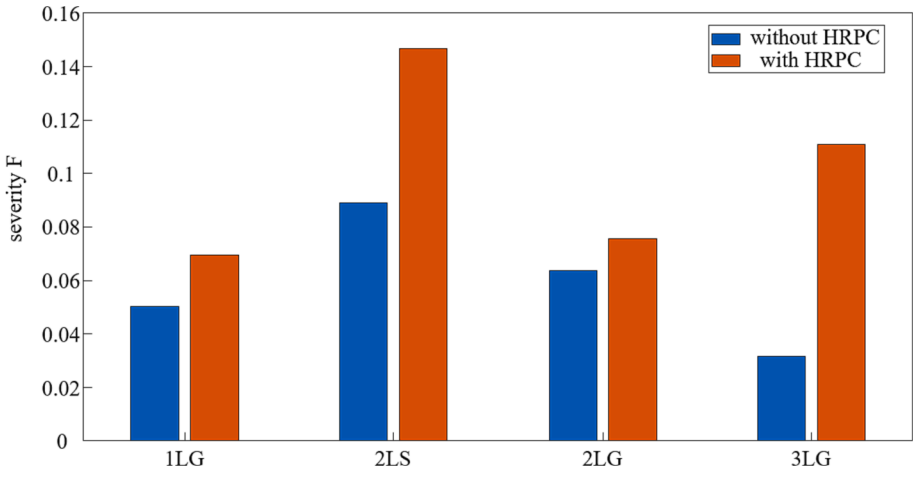


Fig. 15. Comparison of relative severities before and after setting HRPC under different faults.

has a higher TRV when the CB interrupts different faults, which affects the breaking capacity of the circuit breaker and may damage the insulation of the transmission system. Therefore, it is important to analyze the TRV of the circuit breaker when different faults occur in the UHV system once HRPC has been set. Figs. 12, 13, 14, and 15 show the drop-out current, TRV, RRRV, and severity of the circuit breaker, respectively, before and after HRPC set under different faults.

As shown in Figs. 12, 12, and 14, the drop-out current, TRV, and RRRV of the circuit breaker increased after HRPC had been set, and those in case of 2LS were much greater than those in case of 1LG, 2LG,

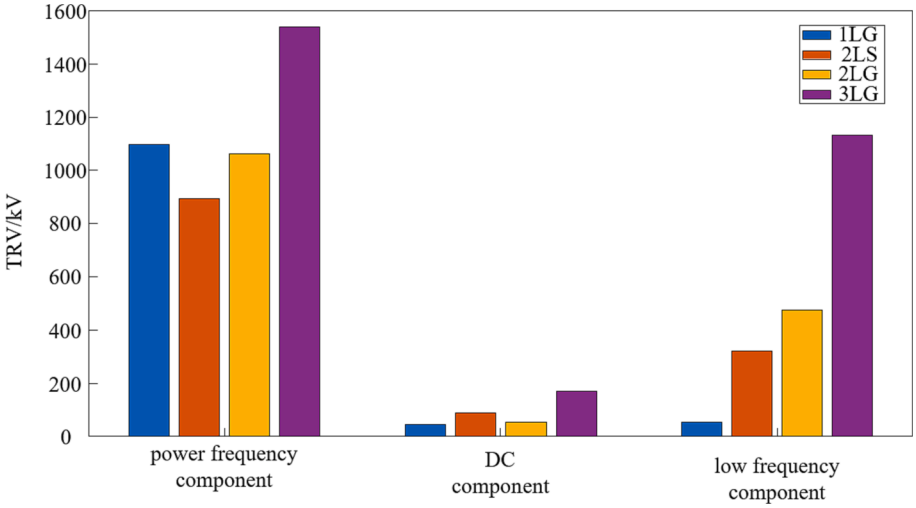


Fig. 16. Values of frequency components of the TRV after Fourier transform.

and 3LG. Owing that 2LG is the final result that 1LG has not ruled out, the drop-out current, TRV, and RRRV of 2LG are very close to those of 1LG.

Fig. 15 shows that the severity of interruption in case of 2LS was much greater than those in case of 1LG, 2LG, and 3LG, where the severity in case of 2LG was close to that in case of 1LG. After installing HRPC, the drop-out current, TRV, and RRRV all increase because the installation of HRPC changes the parameters of the transmission system. The equivalent capacitive reactance of the SC on the line offsets a part of the inductive reactance of the line, resulting in a decrease in the impedance of the line and a higher drop-out current value than the condition without HRPC. When disconnecting the circuit breaker, the SC device will generate high-frequency and low-frequency components, thereby affecting the TRV of the circuit breaker.

The Fourier transform method was used to compare the TRVs of 1LG, 2LG, and 2LS in the UHV transmission system with HRPC to provide a reference for evaluating the interruption performance of the circuit breaker when different faults occur. The value of each frequency component of the TRV in case of a fault, obtained using numerical analysis and simulations, is shown in Fig. 16.

The Fourier transform was also used to analyze the TRVs of 1LG, 2LG, 2LS, and 3LG. The TRV was decomposed into the components of power frequency, DC, and low-frequency components. Fig. 16 shows that the power frequency components and low-frequency components in case of 3LG were both greater than those in case of other faults. The medium- and low-frequency parts of the TRV caused resonance in the coil in the reactor of the transmission system and generated excessive voltage with a higher resonance. A comparison shows that the high-frequency component of 2LS was the largest of all faults considered. It formed a traveling wave inside the GIS and was superimposed on the oscillation to form the most important part of the TRV.

6. Discussion

The 1100 kV Jindongnan Nanyang Jingmen transmission line was selected, and numerical calculation methods were used to study the interruption characteristics. The severity was selected as the indicator of the circuit breaker interrupting characteristics. At the same time, the numerical calculation results were compared with the simulation results to verify the effectiveness which are shown in Fig. 3. Compared with the traditional simulation method to analyse the interrupting characteristics, the numerical calculation is quickly, and can well analyse the influence of line parameters on the interrupting characteristics of circuit breaker.

Secondly, the single-phase ground fault is set as the main research object. The impact of HRPC on the interrupting characteristics of circuit breakers under single-phase ground fault was analysed (as shown in Figs. 4, 5, 6, 7 and 8). The results indicate that when the compensation degree of SC is 0.2, the circuit breaker will be more complicated to interrupt, and the interruption conditions come more severe The compensation level of SCSR has a relatively small effect on the level of disconnection In addition, after the HRPC is set on the UHV transmission circuit, the drop out current, TRV, and RRRV of CB will show a significant increase, which increased the severity of interruption of 1LG and incrementally increased the probability of failure of the CB, The effectiveness of this method was once again verified by comparing the calculated TRV of the circuit breaker in the event of single-phase ground fault on a series compensated transmission line with the experimental results of TRV in existing literature.

Then, the impact of different fault locations and compensation degrees on the interrupting characteristics of the most common faults in the power grid was analysed (as shown in Figs. 9, 10 and 11). The results indicate that, the drop-out current, TRV, RRRV, and the severity of interruption of the circuit breaker increased once HRPC had been set. Moreover, they were considerably higher in case of phase-to-phase faults than the single-phase-to-ground fault, two-phase ground fault, and three-phase ground fault. Owing that two-phase ground faults are the final result that single-phase-to-ground fault has not ruled out, the drop-out current, TRV, and RRRV and the severity level of the two-phase ground fault problem is relatively close to the single-phase-to-ground fault.

Finally, the interrupting characteristics of CB under different fault was compared as shown in Figs. 12, 13, 14 and 15, and Fourier analysis was performed on TRV, as shown in Fig. 16. Values of the power frequency components and low-frequency components in case of a three-phase ground fault were greater than those in case of other faults. The medium- and low-frequency parts of the TRV led to resonance in the coil in the reactor of the transmission system and generated excessive voltage. A comparison showed that the high-frequency component of phase-to-phase faults was the highest of all faults. It formed a traveling wave inside the GIS and was superimposed on the oscillation to form the most important part of the TRV.

According to the calculation results, the optimal setting of HRPC is listed. The corresponding compensation degree of HRPC for the smallest severity is shown in Table 2.

Considering the above results, it can be seen that when a single-phase grounding fault occurs at the midpoint of the line, the optimal setting of HRPC is that the series compensation capacitor is 31.43 μ F. The stepped controlled shunt reactor is 0.44H, at this time, the circuit breaker is the most difficult to interrupt. When phase to phase fault occurs at the midpoint of the line, the optimal setting of HRPC is series compensation capacitor is 47.15 μF. The stepped controlled shunt reactor is 3.94H. When two-phase grounding fault occurs at the midpoint of the line, the optimal setting of HRPC is that the series compensation capacitor is 31.43 μ F. The stepped controlled shunt reactor is 5.13H. When three-phase grounding fault occurs at the midpoint of the line, the optimal setting of HRPC is series compensation 26.94 μ F. The stepped controlled shunt reactor is 5.66H.

7. Conclusions

When the power system has a fault problem, the location where the fault problem occurs and whether hybrid reactive power compensation is set in the system platform, directly affects the interruption parameters of the circuit breaker, including drop-out current, transient recovery voltage, and rate of rise of transient recovery voltage, at the same time, the severe level of interruption can be used to explain the performance of interruption of a circuit breaker for fault at different locations. Therefore, the severity of the interrupting can be used to explain the interrupting performance of the circuit breaker when the fault occurs at each location and with degrees of compensation of hybrid reactive power compensation. The main findings of this topic are as follows:

1) A theoretically equivalent operation model of the single-phase-to-ground fault, three-phase ground fault, phase-to-phase fault, and two-phase ground fault is constructed. the opening and disconnecting current of the phase-to-phase fault problem, and the rate of rise of transient recovery voltage and transient recovery voltage, and two-phase ground fault. When the faults occurred., the effect of compensation level of series compensation on rate of rise of transient recovery voltage and transient recovery voltage is more obvious than the compensation level of stepped controllable shunt reactor.

Table 2
The optimal setting of HRPC for different faults.

	K_c	The value of series compensation capacitor	K_R	The value of stepped controlled shunt reactor
1LG	0.6	31.43μF	0.8	0.44H
2LS	0.4	47.15μF	0.2	3.94H
2LG	0.6	31.43μF	0.1	5.13H
3LG	0.7	26.94μF	0.3	5.66H

2) According to the corresponding accounting results, the relationships among drop-out current, transient recovery voltage, rate of rise of transient recovery voltage and degrees of hybrid reactive power compensation were plotted. They showed that the drop-out current, transient recovery voltage, rate of rise of transient recovery voltage, and the severity of interruption of the circuit breaker increased once hybrid reactive power compensation had been set. Moreover, they were considerably higher in case of phase-to-phase faults than the single-phase-to-ground fault, two-phase ground fault, and three-phase ground fault. Owing that two-phase ground faults are the final result that single-phase-to-ground fault has not ruled out, the drop-out current, transient recovery voltage, and rate of rise of transient recovery voltage and the severity level of the two-connected ground fault problem is relatively close to the single-phase-to-ground fault.
3) The Fourier transform method was used to analyze the transient recovery voltage of single-phase-to-ground fault, two-phase ground fault, phase-to-phase fault, and three-phase ground fault. The transient recovery voltage was decomposed into power frequency components, DC components, and low-frequency components according to frequency. Values of the power frequency components and low-frequency components in case of a three-phase ground fault were greater than those in case of other faults. The medium- and low-frequency parts of the transient recovery voltage led to resonance in the coil in the reactor of the transmission system and generated excessive voltage. A comparison showed that the high-frequency component of phase-to-phase faults was the highest of all faults. It formed a traveling wave inside the GIS and was superimposed on the oscillation to form the most important part of the transient recovery voltage.

CRediT authorship contribution statement

Jingjing Yang: Investigation, Validation, Writing – original draft, Writing – review & editing. **Hongshun Liu:** Conceptualization, Investigation, Supervision, Validation, Writing – review & editing. **Kaining Hou:** Formal analysis. **Zhaoyang Kang:** Data curation. **Qingmin Li:** Supervision. **Qingquan Li:** Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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